Rodolfo's Encyclopedia

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Contents

I Number Theory **5**

1 Basic Theory **7**

1.1 Divisibility 7

1.2 Base Systems 7

1.3 Bézouts' Theorem 7

2 Basic Modular Arithmetic **9**

2.1 Properties 9

2.2 Fermat's Little Theorem 9

2.3 Euler's Theorem 9

3 Arithmetic Funtions **11**

3.1 Number of Divisors 11

3.2 Sum of Divisors 11

3.3 Arithmetic Funtions 11

3.4 Multiplicative Functions 11

3.4.1 Möbius Inversion 11

3.4.2 Dirichlet Convolution 11

3.5 Floor and Ceiling Functions 11

4 Binomial Coefficients **13**

5 Diofantine Equations **15**

5.1 Pythagorean triples 15

5.2 Fermat's Descent 15

5.3 Pell's equation 15

5.4 Vieta's Jump 15

6 Advanced Modular Arithmetic **17**
6.1 Order . . . 17
6.2 Primitive Roots . . . 17
6.3 Fermat's Chrismas Theorem . . . 17
6.4 Thues' Lemma . . . 17
6.5 Chinese Remainde Theorem . . . 17

7 Prime Exponents **19**
7.1 p-adic valuations . . . 19
7.2 LTE . . . 19
7.3 Zsigmondy's Theorem . . . 19

8 Polynomials **21**
8.1 Euclidean Division Lemma . . . 21
8.2 GCD . . . 21
8.3 Vietas Theorem . . . 21
8.4 Irreducibility . . . 21
8.5 Schur's Theorem . . . 21
8.6 Cyclotomic Polynomials . . . 21

9 Quadratic Residues **23**
9.1 Quadratic Reciprocity . . . 23

Part I

Number Theory

Chapter 1

Basic Theory

1.1 Divisibility

Theorem 1. *If $x, y \in \mathbb{Z}$, and $x \mid y$ then $y = 0$ or $\mid x \mid \leq \mid y \mid$*

Theorem 2. *Suppose $c, x, y \in \mathbb{Z}$, and $c \mid x, y$ then $c \mid ax + by$ for any $a, b \in \mathbb{Z}$*

Definition 3.

Theorem 4. *Let a, b, c be three integers. Then*

$$c \mid a, c \mid b, \Rightarrow c \mid \gcd(a, b)$$

Theorem 5.

Theorem 6.

1.2 Base Systems

Definition 7. Given a positive integer b, the base$-b$ representation of a real number r is a string of digits:

$$(a_n a_{n-1} \dots a_1 a_0 . a_{-1} a_{-2} \dots)_b$$

Where $a_i \in \{0, 1, \dots, b-1\}$, and

$$r = a_n b^n + a_{n-1} b^{n-1} \dots a_1 b^1 + a_0 b^0 + a_{-1} b^{-1} a_{-2} b^{-2} \dots$$

Theorem 8. *The base$-b$ representation of any real number is unique.*

1.3 Bézouts' Theorem

Chapter 2

Basic Modular Arithmetic

2.1 Properties

2.2 Fermat's Little Theorem

Theorem 9.

2.3 Euler's Theorem

Chapter 3

Arithmetic Funtions

3.1 Number of Divisors

3.2 Sum of Divisors

3.3 Arithmetic Funtions

3.4 Multiplicative Functions

3.4.1 Möbius Inversion

3.4.2 Dirichlet Convolution

3.5 Floor and Ceiling Functions

Chapter 4

Binomial Coefficients

Definition 10. For non-negative integers a, b, we define

$$\binom{a}{b} = \begin{cases} 0 \text{ if } a < b \\ \dfrac{a!}{b!(a-b)!} \end{cases}$$

Theorem 11 (Lucas Theorem)**.**

Chapter 5

Diofantine Equations

5.1 Pythagorean triples

5.2 Fermat's Descent

5.3 Pell's equation

5.4 Vieta's Jump

Chapter 6

Advanced Modular Arithmetic

6.1 Order

6.2 Primitive Roots

6.3 Fermat's Chrismas Theorem

6.4 Thues' Lemma

6.5 Chinese Remainde Theorem

Chapter 7

Diophantine Equations

7.1 Pythagorean triples

7.2 Fermat's Descent

7.3 Pell's equation

7.4 Vieta's Jump

Chapter 8

Polynomials

8.1 Euclidean Division Lemma

8.2 GCD

8.3 Vietas Theorem

8.4 Irreducibility

8.5 Schur's Theorem

8.6 Cyclotomic Polynomials

Chapter 9

Advanced Modular Arithmetic

9.1 Order

9.2 Primitive Roots

9.3 Fermat's Chrismas Theorem

9.4 Thues' Lemma

9.5 Chinese Remainde Theorem

Chapter 10

Prime Exponents

10.1 p-adic valuations

10.2 LTE

10.3 Zsigmondy's Theorem

Chapter 11

Quadratic Residues

11.1 Quadratic Reciprocity